SignBridge AI: Real-Time Deep Learning-Based American Sign Language Recognition

1. Executive Summary:

Millions of Deaf and hard-of-hearing individuals communicate primarily in American Sign Language (ASL), yet most everyday settings, workplaces, and public services lack real-time interpretation support, creating persistent communication barriers. This project builds SignBridge AI, a deep learning system that recognizes ASL fingerspelling (the 26-letter alphabet) from static images and a curated set of common word-level signs from short video clips, translating recognized signs into on-screen text through a webcam-based Streamlit application.

2. Problem Statement:

Background: ASL is a complete, distinct language with its own grammar and is the primary language for many Deaf individuals in the US, but real-time interpretation tools remain limited, expensive, or dependent on human interpreters not always available on demand. Advances in hand-landmark tracking and sequence-based deep learning make lightweight, accessible sign-recognition tools increasingly feasible.

Objective: To build and deploy a deep learning system that (a) classifies static ASL alphabet gestures from images with high accuracy, and (b) recognizes a curated subset of common word-level ASL signs from short video clips by modeling the motion of hand landmarks over time, presented through a near real-time webcam demo that outputs recognized signs as text.

Scope: The project covers the 26-letter ASL alphabet (plus 'space'/'nothing' classes) for static recognition, and a subset of approximately 25-30 frequently used word-level signs for dynamic, video-based recognition. Full-sentence ASL translation with grammar and non-manual markers (facial expression, body posture) is explicitly out of scope; the project targets isolated sign recognition as an accessible proof of concept rather than a complete ASL-to-English translation system.

3. Data Sources:

Primary Data: N/A

Secondary Data: Sign Language MNIST (Kaggle) for static ASL alphabet images, and a curated subset of the WLASL (Word-Level American Sign Language) video dataset for dynamic word-level signs. Both are publicly available for research and academic use with documented licensing.

4. Methodology:

Data Collection: The alphabet dataset is downloaded directly from Kaggle; the WLASL video subset is downloaded from its public research repository. No manual data collection or in-person signer recording is required.

Data Preparation: For the alphabet dataset, images are normalized and augmented (rotation, brightness, slight scaling) to account for hand-shape and lighting variation. For WLASL videos, hand and pose landmarks are extracted per frame using MediaPipe, converting each video into a normalized sequence of landmark coordinates, which reduces sensitivity to background, skin tone, and camera angle compared to training directly on raw pixels.

Analysis Techniques: A CNN is trained for static alphabet classification from images. For dynamic word-level signs, an LSTM/GRU or temporal CNN is trained on the MediaPipe landmark sequences to capture the motion patterns that distinguish signs, satisfying the project's deep learning requirement through sequence modeling. As a baseline comparison, classical ML models (Random Forest, SVM) are trained on hand-crafted landmark features for the same word-level task to quantify the benefit of the deep sequence model. Evaluation uses accuracy, precision, recall, F1-score, and a confusion matrix to identify commonly confused signs.

Tools: Python, MediaPipe (hand/pose landmark extraction), TensorFlow or PyTorch (CNN and LSTM/GRU models), scikit-learn (classical ML baselines), OpenCV (webcam video capture), Streamlit (real-time demo deployment).

5. Expected Outcomes:

- A trained CNN classifier for the 26-letter ASL alphabet with documented accuracy and F1-score.
- A trained sequence model (LSTM/GRU or temporal CNN) recognizing the chosen subset of word-level ASL signs from landmark sequences.
- A documented comparison of classical ML on hand-crafted landmark features versus the deep sequence model for word-level recognition.
- A deployed Streamlit webcam application that recognizes signs in near real time and displays them as text.
- A written analysis of model limitations, including signer variability, lighting sensitivity, and the constraints of the chosen vocabulary size.

6. Risks and Challenges:

- Variability across signers, camera angles, and video quality in the WLASL dataset. Mitigation: MediaPipe landmark normalization (which is largely invariant to background and skin tone) plus data augmentation on landmark sequences.
- Real-time performance constraints for a webcam-based demo. Mitigation: using lightweight landmark-based sequence models rather than training directly on raw video frames, which keeps inference fast enough for near real-time use.
- Limited vocabulary and lack of full-sentence grammar support. This is explicitly scoped and documented as an isolated-sign recognition proof of concept, not presented as a complete ASL translation system.
- Webcam demo performance depends on lighting and hand visibility. Mitigation: on-screen guidance in the app for hand positioning and lighting, with graceful handling of low-confidence predictions.

7. Conclusion:

SignBridge AI applies deep learning to a meaningful accessibility problem, combining image classification for ASL fingerspelling with sequence modeling for word-level sign recognition. By grounding the dynamic-sign component in MediaPipe landmark sequences rather than raw video, the project keeps training and inference computationally feasible while still requiring a genuine deep learning component (LSTM/GRU or temporal CNN) to capture motion over time. The scope is deliberately bounded to isolated alphabet and word-level recognition on public datasets, keeping the project realistic and achievable within the available timeline while still addressing a real, underserved accessibility need.